

Milestone Progress Evaluation

Milestone 1

1. Project title, names, and email addresses of team members

Basic CPS Attack and Defense testbed

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2. Faculty Advisor

Dr. Sneha Sudhakaran - ssudhakaran@fit.edu

3. Client or user

Dr. Sneha Sudhakaran - Department of Electrical Engineering and Computer Science at Florida Institute of Technology

4. Progress of current Milestone

Task	Completion %	Karla	Zach	To do
1. Compare and Select Technical Tools	90%	50%	50%	Finish getting Siemens Step 7
2. "Hello World" demos	100%	50%	50%	None
3. Resolve Technical Challenges	100%	50%	50%	None
4. Design Document	100%	50%	50%	None
5. Requirements Document	100%	75%	25%	None
6. Test Document	100%	25%	75%	None

5. Discussion of each accomplished task and obstacles

Task 1: This was fairly straightforward. We just needed to pick what tools for scripting, analysis, logging & PLC programming. For PLC programming, each of the PLC manufacturers is designed to use only one software set. Of the three we are working with, only Modicon was able to get the software on day 1. Allen-Bradley's software is free but has restricted access. Since then, we have gotten & set up this software. Siemens has restricted access and is normally paid. We passed this along to our advisor, who is going through official channels to see if we can get this Siemens software for free/heavily discounted.

Task 2: This task has been completed. We were able to set up the connection between the PLCs and the historian via the switch and detect the presence of each PLC. To verify PLC communication over the switch, we used software to connect to the PLC and ran a simple hello world demo. For example, we were able to connect the Allen Bradley PLC to the trainer, and using the Rockwell software, we were able to show a simple demonstration of the PLC communicating with the software via the input and the output to show a light turning on and off when the input is turned on and off using the programmed software. One obstacle we faced was with the Siemens PLC: the software, Step 7, requires the school to request access, so the connection between the PLC and the software will be delayed until we can obtain this access.

Task 3: We were able to understand how the PLC is connected to the trainer via inputs and outputs, as well as how the sensors connected to each PLC would communicate with the software and exchange information with these PLCs.

Task 4: This task has been completed. We have made a 6-page design plan to the specifications given. It has been linked on the project site. The only major obstacle was ensuring it was fully up to specification and included all relevant details. This was overcome with persistence in the task & chipping away at parts of the document until complete.

Task 5: This task has been completed. We have prepared a 5-page requirements document based on the specifications given. It has been linked on the project site. The only major obstacle was ensuring it was fully up to specification and included all relevant details. This was overcome with persistence in the task & chipping away at parts of the document until complete.

Task 6: This task has been completed. We have made a 5-page test document to the specifications given. It has been linked on the project site. The only major obstacle was ensuring it was fully up to specification and included all relevant details. This was overcome with persistence in the task & chipping away at parts of the document until complete.

6. Contribution of each team member to the current Milestone

Karla Daley: set up the connection between the PLCs and the trainer and programmed the software so that the program could be uploaded to the PLC. This was successfully completed for the Allen-Bradley PLC but not for the Siemens PLC. Each input and output was tested to verify the connection between the PLC and the trainer. For the “Hello World” demo using the PLC, trainer, and software, the Allen-Bradley PLC successfully demonstrated communication between the input and output. When Input 1 was turned on, the light connected to Output 1 turned on, and when Input 1 was turned off, the Output 1 light also turned off. I also created the wiring needed to connect the PLCs and trainer using 22 AWG wire, ferrules, and a crimper. Enough wiring was prepared for both the positive and negative power connections. To organize the power distribution, I created separate positive and negative groups using the DIN rail terminal blocks and jumper bars. I also researched and documented the input-output mapping and connectivity between the PLCs and the trainer. This helped organize the hardware connections and made it easier to identify which trainer inputs and outputs corresponded to the PLC terminals when programming the system in software.

Zach Lavoie: Reflashed the computer that will be used as the historian and set up Windows. I also reconfigured the Ethernet drivers after they were deactivated during the Windows installation. Set up the first network switch using the TP-Link configuration software and documented the passwords for both the historian and the first network switch. I also installed the Allen-Bradley PLC software along with the required .NET dependency.

7. Plan for the next Milestone

Task	Karla	Zach
1. Demo internal network attacks	Set up PLCs so that when under attack, we can observe the physical outcome.	Reflash the second computer & make it able to see the network PLCs.
2. Set up sensor connection to all 3 PLCs	Set up the sensors that the PLCs will connect to. Make sure these also affect the PLC trainer readout.	Allow the historian to see all sensor data & create a logging system for it.
3. Finish setting up the Siemens PLC	Add Siemens PLC to the current hardware testbed setup, including software.	Acquire & setup Siemens Step 7 to interface with the PLCs through the switch.

8. Discussion of each planned task for the next Milestone

Task 1: The goal is to have a demonstration of an internal network attack ready to present. The work will be split, where Karla sets up the PLC trainer dash for observable data output. Zach will set up the second computer to then become an on-network attacker that can run an attack against the connected PLCs. This can be checked by running a Python script on the HMI attack computer & there should be an observable outcome on the PLC trainer.

Task 2: The goal is to set up sensors that connect to the PLC, which then passes this data to the historian. Karla will set up the sensors & connect them to the PLCs & the trainer. This will be set up so that the sensor data can be viewed in the historian & have some impact on the PLC trainer. Zach will create a basic logging system on the historian to gather all the PLC data & make it an easily exportable & reviewable dataset.

Task 3: This was the only part of Milestone 1 that could not be fully completed due to this software's restricted nature. The goal is just to finish acquiring Siemens Step 7 software & set up the Siemens PLCs with it. Zach, once given the software by the advisor, will set it up on the historian & ensure PLC access through the network switch. Karla will add the PLCs to the physical testbed so they can be programmed and used like the other 2 PLC brands with sensors & readouts.

9. Date(s) of meeting(s) with Client during the current milestone

09/18/2026

09/22/2026

09/24/2026

10. Client feedback on the current milestone

See Faculty Advisor Feedback below.

11. Date(s) of meeting(s) with Faculty Advisor during the current milestone

09/18/2026

09/22/2026

09/24/2026

12. Faculty Advisor feedback on each task for the current Milestone

Task 1: [Advisor feedback] Completed waiting for Seimens license

Task 2: [Advisor feedback] Completed

Task 3: [Advisor feedback] Completed

Task 4: [Advisor feedback] Completed

Task 5: [Advisor feedback] Completed

Faculty Advisor Signature: _____ Sneha Sudhakaran _____ Date: _09/27 2026 _____